

# A 3-Sphere with One Imaginary Axis: Two Curvature Signs and an Equatorial $S^2$ , Not a Selection of de Sitter

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## Abstract

The virtual modes of this program are to live on a 3-sphere whose fourth embedding axis is imaginary and whose values are curvature, so the same sphere has de Sitter and anti-de Sitter regions. That habitat is author ontology. The geometry underneath it is not.

Round  $S^3$  of radius  $\rho$  has Ricci scalar  $6/\rho^2$ , computed from the metric.  $\rho \rightarrow i\ell$  sends  $\mathcal{R} \rightarrow -6/\ell^2$ . One family, two signs. The equator is a round  $S^2$ . The Haar mean of the embedding coordinate  $X_4 = \cos \chi$  is identically zero. All four identities **[P]**.

Identifying  $X_4$  with curvature, and putting virtual particles on this  $S^3$ , remains a conjecture. It does not copy FGHMV onto de Sitter:  $\dim \mathfrak{so}(1,4) = 10 < 15 = \dim \mathfrak{so}(2,4)$  is still Paper 17. It does not measure  $\Lambda$ .

## 1 The 3-geometry

$$ds^2 = \rho^2(d\chi^2 + \sin^2 \chi d\Omega_2^2).$$

Christoffel reduction of this metric gives  $\mathcal{R} = 6/\rho^2$ . The substitution  $\rho = i\ell$  is algebraic and flips the sign. That is the spherical / hyperbolic pair:  $S^3$  and  $H^3$ . Calling the two signs dS and AdS *areas* is allowed as a name for the two 3-geometries. They are not yet 4d de Sitter and anti-de Sitter spacetimes.

The equator  $\chi = \pi/2$  is  $S^2$ . Papers 23–27 already work on a 2-sphere topology (the diamond waist). That is a kinship of topology, not a derivation of those lattice results from this embedding.

## 2 Mean zero

On the unit  $S^3$ ,  $\langle \cos \chi \rangle = 0$  by oddness of the Haar integrand. If  $X_4$  is curvature, the uniform mean is zero. That is the foam picture I want, *given* the identification. It is not a derivation of the identification, and it does not select  $\Lambda > 0$ . A mean-zero axis is Minkowski as the average, with both signs as fluctuations.

### 3 What this does not move

Paper 17 stands. Paper 19 still says the spinless Einstein+ $\Lambda > 0$  vacuum is de Sitter, as a metric identity. Paper 27 still says the CHM envelope beats a flat kernel on the *cut*, with weak tracking. This note adds a home for the virtual modes. It does not finish the selection.

**Admission.** *I wrote the 3-sphere I have used as the habitat of virtual modes. I derived the Wick sign and the Haar mean. I did not write that the modes were derived, and I did not write that de Sitter was selected.*